

Measuring the External Field: A Cortical-Proxy Content Observable and the Mean-Field Bound on Media-Driven Opinion Change

Brian Mwai, 1050555, B.Sc. Physics, Department of Natural Sciences, The Catholic University of Eastern Africa. Supervisor: Dr. Songa Mutambi.

Abstract

Media is routinely modelled in statistical physics as an external field acting on a population of coupled opinion states, and that field is almost never measured. This dissertation builds an instrument that measures a candidate observable from content itself, applies it to a corpus, and reports what the measurement can and cannot support.

The observable is a cortical proxy. Text is spoken, transcribed, embedded, and passed to TRIBE v2, a released encoder that predicts cortical response; two networks are averaged from that prediction, one associated with affective salience and one with deliberative control, and the index is their signed difference. No participant was scanned, and the observable is never identified with any subcortical structure.

A corpus of 400 items across four categories, matched on length, was scanned on a GPU. The categories separate at $\eta^2 = 0.1068$, $p = 1.035 \times 10^{-9}$, against a design powered to detect $\eta^2 = 0.0268$. The index discriminates the pre-scan manipulative label at $AUC = 0.6274$, against 0.9758 for a TF-IDF baseline on the same rows, so it is reported as a field observable and not as a classifier. The separation is carried by fear-activating content; high-outrage content raises both networks almost equally and does not separate. Each category is drawn from a single source dataset, so the separation cannot be attributed to framing rather than provenance. A second scanning session gives test-retest $ICC = 0.8725$ and replicates the separation, so the instrument supports category-level claims and no per-item verdict.

In the mean-field treatment media enters as a field $h = \alpha X$ on the measured observable. Calibrating α against the corpus does not identify it, so no value is quoted. What survives is a necessary condition, $\alpha \geq h_c(\beta J)/\Delta X$: the measured spread $\Delta X = 0.1241$ gives $\alpha \geq 4.2938$ at $\beta J = 2$, a constraint on any future proposal of this mechanism within the mean-field model.

Keywords: sociophysics, opinion dynamics, mean-field Ising model, external field, encoding models.